

6. Series A: the noise plate

This series captures the 9 stations of the ladder plus the 2 legacy depths (700 and 1000 mm) at 5 field positions each, 100 frames per pose, plus a tilt sub-series. The station order is randomized, with a logged seed, so that slow drift cannot masquerade as a Z dependence. The planner has already done this: follow the order column.

Mount T2. Do the mount check (Step 4.8). The stations are every station of the ladder from Z_MIN (400 mm) to Z_MAX (1600 mm), 400, 476, 566, 673, 800, 951, 1131, 1345, 1600 mm, and the legacy depths 700 and 1000 mm. The five field positions are the center and four corners at 0.6 of the half field, all fronto-parallel (Figure 5).

Check that the plate fully covers the analysis region at every station. At Z_MIN off-axis this may limit the field offset. The planner has pulled such poses inward; see `plan_summary.txt`.

Before the first station, capture a drift sentinel: center, Z = 800 mm, 30 frames. The plan repeats the sentinel every 60 minutes and after the last station.

At each station: move, wait the settle time, then capture the frames of the row. Log the read-back robot pose, the sensor temperature, and the timestamps.

Tilt sub-series. At the center and at the 3 reduced stations (400, 800, 1600 mm), T2 is tilted about V, then about H, through each angle of 0, 15, 30, 45 degrees. Capture 50 frames per pose. Incidence angle changes both the per-pixel noise and the fill rate. These rows have sub-series tilt.

Repeat-mount check. Dismount T2, re-mount it, and repeat the Z = 800 mm center station (sub-series remount). The difference shows how much of the bias comes from re-mounting the target; the adapter repeats to 0.02 mm.

Do not delete any frames. Tell the engineer if a pose shows a warning in §12.

Engineer: if Step 4.2 calls for it, repeat the series with the sensor's filters off. This repeat is outside the capture budget of §13. Plan it with `--filters-off` and record the new configuration in `sensor_config.json` with a new `config_id`.